

WIRELESS LINK BUDGET ANALYSIS REPORT

Course: 3171608 - Wireless Communication (WC) | Micro-Project

Scenario: Wi-Fi 6 (802.11ax) 2.4 GHz Indoor/Campus | Generated: 2026-10-05 23:29:38 | Model: Log-Distance Model

LINK STATUS: X FAIL / MARGIN DEFICIT — LINK DOWN / FAILED (Margin Deficit: 7.27 dB)
Received Power: -89.27 dBm | Sensitivity: -82.00 dBm | Fade / Link Margin: -7.27 dB

Executive Performance Metrics

EIRP	22.50 dBm (0.178 W)	Received Power (Prx)	-89.27 dBm (1.18 pW)
Free Space Path Loss	78.24 dB	Model Path Loss	97.27 dB
Total Environmental Loss	16.00 dB	Total Attenuation	114.27 dB
Thermal Noise Floor	-94.96 dBm	Signal-to-Noise Ratio (SNR)	5.70 dB
Max Achievable Distance	0.046 km	Shannon Capacity	44.72 Mbps
Max Allowable Path Loss	90.00 dB	Carrier Wavelength (λ)	0.1230 m (12.30 cm)

System Input Parameters

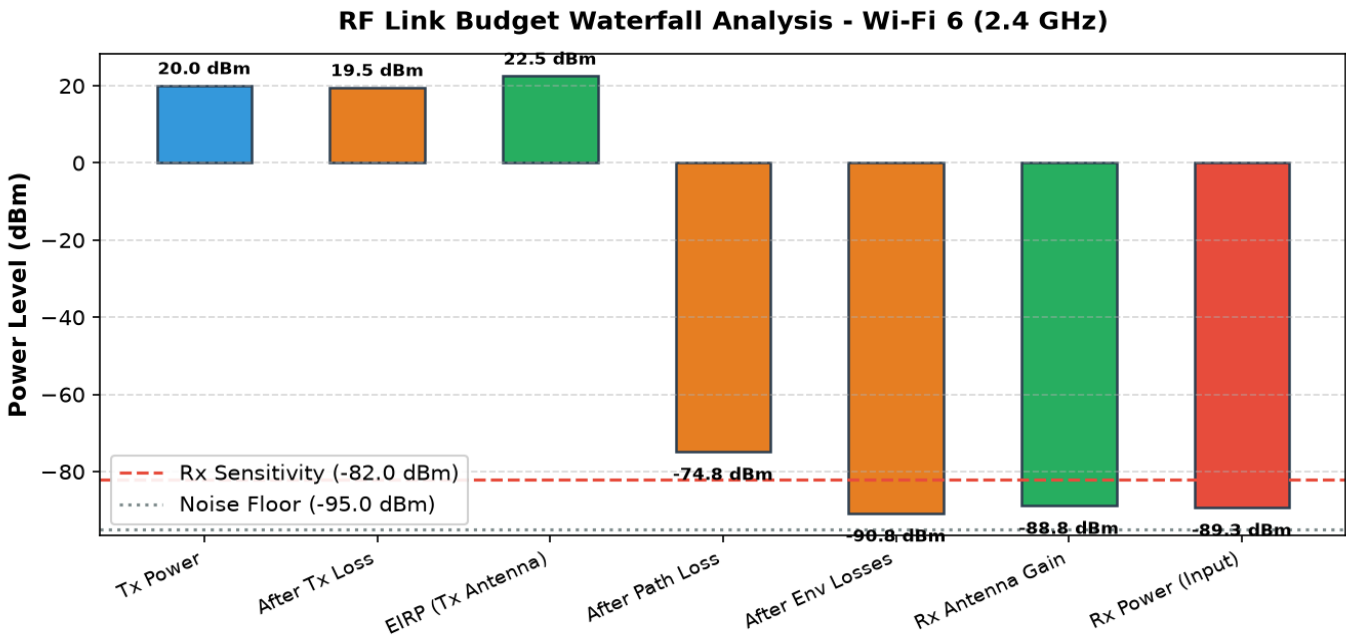
Subsystem	Parameter Name	Value	Units / Notes
Transmitter (Tx)	Tx Output Power (Ptx)	20.00	dBm
	Tx Antenna Gain (Gtx)	3.00	dBi
	Tx Cable & Connector Losses	0.50	dB
	Tx Antenna Height (hb)	30.0	meters
Channel / Path	Carrier Frequency (f)	2437.00	MHz
	Link Distance (d)	0.080	km
	Propagation Model	Log-Distance Model	Empirical/Analytical
	Path Loss Exponent (n)	3.00	Dimensionless
	Rain Attenuation	0.00	dB
	Atmospheric & Gas Loss	0.00	dB
	Obstacle / Penetration Loss	6.00	dB
Receiver (Rx)	Fade Margin Allocation	10.00	dB
	Rx Antenna Gain (Grx)	2.00	dBi
	Rx Cable & Filter Loss	0.50	dB
	Rx Sensitivity Threshold	-82.00	dBm
	Front-End Noise Figure (NF)	6.00	dB

	Channel Bandwidth (B)	20.00	MHz
	Noise Temperature (T)	290.0	Kelvin (K)

Link Budget Stage-by-Stage Power Accounting

Stage Description	Incremental Gain/Loss (dB)	Cumulative Power Level (dBm)
Tx Power	+20.00 dB	20.00 dBm
After Tx Loss	-0.50 dB	19.50 dBm
EIRP (Tx Antenna)	+3.00 dB	22.50 dBm
After Path Loss	-97.27 dB	-74.77 dBm
After Env Losses	-16.00 dB	-90.77 dBm
Rx Antenna Gain	+2.00 dB	-88.77 dBm
Rx Power (Input)	-0.50 dB	-89.27 dBm

RF Link Budget Visual Plots



Core Mathematical Formulas Applied:

- Friis Received Power: $P_{rx} \text{ (dBm)} = P_{tx} + G_{tx} - L_{tx} - PL + G_{rx} - L_{rx} - L_{env}$
- Free Space Path Loss: $FSPL \text{ (dB)} = 32.44 + 20 \cdot \log_{10}(d_{km}) + 20 \cdot \log_{10}(f_{MHz})$
- Thermal Noise Floor: $N \text{ (dBm)} = -174 \text{ dBm/Hz} + 10 \cdot \log_{10}(B_{Hz}) + NF \text{ (dB)}$
- Link Margin: $\text{Margin (dB)} = P_{rx} \text{ (dBm)} - P_{rx,sens} \text{ (dBm)}$
- Shannon Capacity: $C = B \cdot \log_2(1 + SNR_{linear}) \text{ (bps)}$